

COP2800C Module 5 Graded Programming Assignment

I recommend that you use the Horizon system for this assignment unless you are comfortable working with environment variables and execution search paths. All necessary configurations are already in place on the Horizon system.

Make the following changes to the Penguin class and PenguinRookery class from Module 4. Use the Module 5 Practice Exercise as a reference.

- Add an equals method to the Penguin class. The rule to determine whether two penguins are equal are if they are of the same species. **You must use a ternary operator** in the equals method to **verify that the parameter is an instance of a Penguin** and to **compare the species attribute of both objects with the == operator** (you will need to cast the parameter object to a Penguin and use the correct parentheses syntax for this to work correctly).
- For the existing array references in the PenguinRookery class (in the toString method and main method), **use an Integer wrapper object and postfix increment operators** instead of numeric literals for the array index values. **Do not use loops! We have not covered them in class yet.**
- Add code to the main method in the PenguinRookery class to instantiate a test Penguin object; then choose the array element which is equal to your test Penguin (has the same species) to **compare the results of comparing using the == operator and the equals method.**

Create a JAR file for your PenguinRookery application which contains your source code and class files for all three classes.

Attach and submit the following artifacts:

- Your JAR file (you do not need to submit your source code files since your JAR file will contain them)
- One or more screen snips which show successful compilation of your code and execution of the JAR file from the command line as demonstrated in the practice exercise
 - **Do not submit screen snips from jGrasp**, your submissions must show only command line operations.

Expected output is as follows:

```
Our rookery contains the following penguins:
First penguin is a(n) CHINSTRAP.
Second penguin is a(n) GENTOO.
And the final penguin is a ADELIE.
There are a total of 3 penguin species in our rookery.
Testing mutators and accessors
first penguin: ADELIE
second penguin: CHINSTRAP
third penguin: GENTOO
compare Penguins with ==, result is false
compare Penguins with equals method, result is true
```